Problem Set-4

Question 1.
Let X and Y be two topological spaces; and $a \in X,\ b \in Y$ be two given points. Prove that the maps

$$f : X \to X \times Y \quad \text{and} \quad g : Y \to X \times Y$$

given by

$$f(x) = (x, b) \quad \text{and} \quad g(y) = (a, y)$$

are embeddings.

Proof.
Recall that a map $h : Z \to W$ is an embedding if it is a homeomorphism from Z onto $h(Z)$, equipped with the subspace topology.

1. The map $f : X \to X \times Y$, $f(x) = (x, b)$.
Step 1: f is continuous.
Let $U \times V$ be an arbitrary basic open set in $X \times Y$, where $U \subseteq X$ and $V \subseteq Y$ are open.
Then

$$\begin{aligned} f^{-1}(U \times V) &= \{x \in X : f(x) \in U \times V\} \\ &= \{x \in X : (x, b) \in U \times V\} \\ &= \begin{cases} U, & b \in V, \\ \varnothing, & b \notin V. \end{cases} \end{aligned}$$

In either case, $f^{-1}(U \times V)$ is open in X. Hence f is continuous.

Step 2: f is injective.
Suppose that $f(x_1) = f(x_2)$. Then

$$(x_1, b) = (x_2, b).$$

Therefore, $x_1 = x_2$, so f is injective.

Step 3: The inverse map $f^{-1} : f(X) \to X$ is continuous.
Observe that

$$f(X) = X \times \{b\}.$$

Consider the first projection map

$$\pi_X : X \times Y \to X, \qquad \pi_X(x, y) = x.$$

Since projection maps are continuous, its restriction

$$\pi_X|_{f(X)} : f(X) \to X$$

is continuous.

But

$$\pi_X(f(x)) = \pi_X(x, b) = x.$$

Thus

$$f^{-1} = \pi_X|_{f(X)}.$$

Hence f^{-1} is continuous.

Therefore, f is a homeomorphism from X onto $f(X)$, and consequently f is an embedding.

2. The map $g : Y \to X \times Y$, $g(y) = (a, y)$.

Step 1: g is continuous.

Let $U \times V$ be an arbitrary basic open set in $X \times Y$.

Then

$$\begin{aligned} g^{-1}(U \times V) &= \{y \in Y : g(y) \in U \times V\} \\ &= \{y \in Y : (a, y) \in U \times V\} \\ &= \begin{cases} V, & a \in U, \\ \varnothing, & a \notin U. \end{cases} \end{aligned}$$

In either case, $g^{-1}(U \times V)$ is open in Y. Hence g is continuous.

Step 2: g is injective.

Suppose that $g(y_1) = g(y_2)$. Then

$$(a, y_1) = (a, y_2).$$

Therefore, $y_1 = y_2$, so g is injective.

Step 3: The inverse map $g^{-1} : g(Y) \to Y$ is continuous.

Observe that

$$g(Y) = \{a\} \times Y.$$

Consider the second projection map

$$\pi_Y : X \times Y \to Y, \qquad \pi_Y(x, y) = y.$$

Since projection maps are continuous, its restriction

$$\pi_Y|_{g(Y)} : g(Y) \to Y$$

is continuous.

But

$$\pi_Y(g(y)) = \pi_Y(a, y) = y.$$

Thus

$$g^{-1} = \pi_Y|_{g(Y)}.$$

Hence g^{-1} is continuous.

Therefore, g is a homeomorphism from Y onto $g(Y)$, and consequently g is an embedding.

Hence both f and g are embeddings.

Question 2.

Suppose that $f, g : X \to \mathbb{R}$ are two continuous maps.

(i) Prove that the set

$$\{x \in X \mid f(x) \leq g(x)\}$$

is closed in X.

(ii) Prove that the map $h : X \to \mathbb{R}$ given by

$$h(x) = \min\{f(x), g(x)\}$$

is continuous.

Proof.

(i) The set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in X.

Let

$$A = \{x \in X \mid f(x) \leq g(x)\}.$$

We know that f and g are continuous. Since the subtraction map

$$\varphi : \mathbb{R} \times \mathbb{R} \to \mathbb{R}, \qquad \varphi(u, v) = u - v$$

is continuous, the map

$$F : X \to \mathbb{R}, \qquad F(x) = f(x) - g(x)$$

is continuous.

Now observe that

$$\begin{aligned} A &= \{x \in X \mid f(x) \leq g(x)\} \\ &= \{x \in X \mid f(x) - g(x) \leq 0\} \\ &= F^{-1}((-\infty, 0]). \end{aligned}$$

Since $(-\infty, 0]$ is closed in $\mathbb{R}$ and F is continuous, its inverse image is closed in X.

Therefore,

$\{x \in X \mid f(x) \leq g(x)\}$ is closed in X.

(ii) The map $h : X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.

We use the identity

$$\min\{a, b\} = \frac{a + b - |a - b|}{2}, \qquad a, b \in \mathbb{R}.$$

Therefore, for every $x \in X$,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}.$$

Since f and g are continuous, the maps

$$f + g : X \to \mathbb{R}$$

and

$$f - g : X \to \mathbb{R}$$

are continuous.

Also, the absolute value map

$$|\cdot| : \mathbb{R} \to \mathbb{R}, \qquad t \mapsto |t|$$

is continuous.

Hence the composition

$$x \mapsto |f(x) - g(x)|$$

is continuous.

Consequently,

$$h(x) = \frac{f(x) + g(x) - |f(x) - g(x)|}{2}$$

is continuous, since sums, differences and scalar multiples of continuous real-valued functions are continuous.

Thus,

$$\boxed{h : X \to \mathbb{R} \text{ is continuous.}}$$

Question 3.

Let $\{A_\alpha\}$ be a collection of subsets of X such that

$$X = \bigcup_\alpha A_\alpha.$$

Suppose that $f : X \to Y$ is a map such that the restriction map

$$f|_{A_\alpha} : A_\alpha \to Y$$

is continuous for each α.

(i) Prove that if the collection $\{A_\alpha\}$ is finite and each set A_α is closed, then f is continuous.

(ii) Find an example where the collection $\{A_\alpha\}$ is countable and each set A_α is closed, but f is not continuous.

(iii) An indexed family of sets $\{A_\alpha\}$ is said to be locally finite if each point x has a neighbourhood that intersects A_α for finitely many values of α. Prove that if the family $\{A_\alpha\}$ is locally finite and each set A_α is closed, then f is continuous.

Recall.

A map $f : X \to Y$ is continuous if and only if the preimage of every closed subset of Y is closed in X.

Also, if A is a closed subset of X, then every relatively closed subset of A is closed in X.

(i) The collection $\{A_\alpha\}$ is finite and each A_α is closed.

Proof.

Suppose

$$X = A_1 \cup A_2 \cup \cdots \cup A_n,$$

where each A_i is closed in X.

We are given that the restriction

$$f|_{A_i} : A_i \to Y$$

is continuous for every $i = 1, 2, \ldots, n$.

We need to prove that $f : X \to Y$ is continuous.

Let C be an arbitrary closed subset of Y.

Consider

$$f^{-1}(C) = \{x \in X : f(x) \in C\}.$$

Since $X = \bigcup_{i=1}^{n} A_i$, we can write

$$f^{-1}(C) = \bigcup_{i=1}^{n} \left(f^{-1}(C) \cap A_i\right).$$

Now observe that

$$f^{-1}(C) \cap A_i = (f|_{A_i})^{-1}(C).$$

Since $f|_{A_i}$ is continuous and C is closed in Y,

$$(f|_{A_i})^{-1}(C)$$

is closed in A_i.

But A_i itself is closed in X. Therefore, a relatively closed subset of A_i is also closed in X.

Thus,

$$f^{-1}(C) \cap A_i$$

is closed in X for every i.

Consequently,

$$f^{-1}(C) = \bigcup_{i=1}^{n} \left(f^{-1}(C) \cap A_i\right)$$

is a finite union of closed subsets of X.

Hence, $f^{-1}(C)$ is closed in X.

Since C was an arbitrary closed subset of Y, we conclude that

$$\boxed{f : X \to Y \text{ is continuous.}}$$

□

(ii) Find a countable collection of closed sets covering X for which f is not continuous.

Example.

Take

$$X = \mathbb{R}, \qquad Y = \mathbb{R}.$$

Define

$$A_0 = \{0\}$$

and, for $n \in \mathbb{N}$,

$$A_n = \left[-n, -\frac{1}{n}\right] \cup \left[\frac{1}{n}, n\right].$$

Step 1. Show that each A_n is closed.

Clearly, $A_0 = \{0\}$ is closed in $\mathbb{R}$.

Also, each A_n is a finite union of closed intervals. Hence, every A_n is closed in $\mathbb{R}$.

Step 2. Show that the collection is countable and covers $\mathbb{R}$.

The collection

$$\{A_0, A_1, A_2, \ldots\}$$

is countable.

Moreover, every nonzero real number belongs to some A_n.

Indeed, given $x \neq 0$, choose n sufficiently large such that

$$n \geq |x| \qquad \text{and} \qquad \frac{1}{n} \leq |x|.$$

Then $x \in A_n$.

Since $0 \in A_0$, we obtain

$$\boxed{\mathbb{R} = \bigcup_{n=0}^{\infty} A_n.}$$

Step 3. Define the map f.

Define

$$f : \mathbb{R} \to \mathbb{R}$$

by

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

We now show that the restriction of f to every A_n is continuous.

- On $A_0 = \{0\}$, the restriction is the constant function $f(0) = 1$, which is continuous.
- On every A_n, $n \geq 1$, we have $0 \notin A_n$. Therefore, $f|_{A_n}$ is the constant zero function, which is continuous.

Thus, every restriction

$$f|_{A_n} : A_n \to \mathbb{R}$$

is continuous.

Step 4. Show that f is not continuous.

Consider the open set

$$U = \left(\frac{1}{2}, \frac{3}{2}\right) \subseteq \mathbb{R}.$$

We have

$$f^{-1}(U) = \{0\}.$$

But $\{0\}$ is not open in $\mathbb{R}$.

Therefore, f is not continuous.

Hence, we have constructed a countable closed cover of $\mathbb{R}$ such that every restriction is continuous, but the original map is not continuous.

Thus,

The conclusion of (i) need not hold for a countable cover.

□

(iii) Prove the result when $\{A_\alpha\}$ is locally finite and each A_α is closed.

Proof.

We are given that:

1. $X = \bigcup_\alpha A_\alpha$.
2. Every A_α is closed in X.
3. The family $\{A_\alpha\}$ is locally finite.
4. Every restriction $f|_{A_\alpha} : A_\alpha \to Y$ is continuous.

We need to prove that $f : X \to Y$ is continuous.

We will use the following important result.

Lemma. The union of a locally finite family of closed subsets of a topological space is closed.

Proof of the lemma.

Let $\{F_\alpha\}$ be a locally finite family of closed subsets of X.

We want to show that

$$F = \bigcup_\alpha F_\alpha$$

is closed in X.

Take an arbitrary point $x \in X \setminus F$.

Since the family is locally finite, there exists an open neighbourhood U of x that intersects only finitely many members of the family.

Suppose these members are

$$F_{\alpha_1}, F_{\alpha_2}, \ldots, F_{\alpha_n}.$$

Now, $x \notin F$, so in particular,

$$x \notin F_{\alpha_i}, \qquad i = 1, \ldots, n.$$

Since each F_{α_i} is closed, its complement is open. Thus,

$$X \setminus F_{\alpha_i}$$

is an open neighbourhood of x.

Consider

$$V = U \cap \bigcap_{i=1}^{n} (X \setminus F_{\alpha_i}).$$

This is an open neighbourhood of x, being a finite intersection of open sets.

We claim that

$$V \cap F = \varnothing.$$

Indeed:

- For the finitely many sets $F_{\alpha_1}, \ldots, F_{\alpha_n}$, the set V does not intersect any of them.
- For every other index α, U, and hence V, does not intersect F_α, by the choice of U.

Therefore, V does not intersect their union F.

So every point in $X \setminus F$ has an open neighbourhood contained in $X \setminus F$.

Hence, $X \setminus F$ is open, which means

$$\boxed{\bigcup_\alpha F_\alpha \text{ is closed in } X.}$$

This proves the lemma.

Now we prove the required result.
Let C be an arbitrary closed subset of Y.
Consider

$$f^{-1}(C).$$

Since $X = \bigcup_\alpha A_\alpha$, we have

$$\begin{aligned} f^{-1}(C) &= \bigcup_\alpha \left(f^{-1}(C) \cap A_\alpha\right) \\ &= \bigcup_\alpha (f|_{A_\alpha})^{-1}(C). \end{aligned}$$

Now, since every restriction $f|_{A_\alpha}$ is continuous and C is closed in Y,

$$(f|_{A_\alpha})^{-1}(C)$$

is closed in A_α.
Since A_α is closed in X, it follows that

$$F_\alpha := (f|_{A_\alpha})^{-1}(C)$$

is closed in X.
Furthermore, since

$$F_\alpha \subseteq A_\alpha$$

and the family $\{A_\alpha\}$ is locally finite, the family $\{F_\alpha\}$ is also locally finite.
Therefore, by our lemma, the union

$$f^{-1}(C) = \bigcup_\alpha F_\alpha$$

is closed in X.
Since C was an arbitrary closed subset of Y, the preimage of every closed subset of Y is closed in X.
Consequently,

$$\boxed{f : X \to Y \text{ is continuous.}}$$

□

Question 4.
Let $f : X \to A$ and $g : Y \to B$ be two continuous maps. Define

$$f \times g : X \times Y \to A \times B$$

by

$$(f \times g)(x, y) = (f(x), g(y)).$$

Prove that $f \times g$ is a continuous map.

Proof.
We are given that

$$f : X \to A \qquad \text{and} \qquad g : Y \to B$$

are continuous maps.

We need to prove that

$$f \times g : X \times Y \to A \times B$$

is continuous.

Recall that the product topology on $A \times B$ has a basis consisting of sets of the form

$$U \times V,$$

where U is open in A and V is open in B.

We will use the characterization of continuity in terms of the preimages of basic open sets.

Step 1. Consider an arbitrary basic open set in $A \times B$.
Let

$$U \times V$$

be an arbitrary basic open set in $A \times B$, where $U \subseteq A$ and $V \subseteq B$ are open.

We have

$$\begin{aligned}
&(f \times g)^{-1}(U \times V) \\
&= \{(x, y) \in X \times Y : (f \times g)(x, y) \in U \times V\} \\
&= \{(x, y) \in X \times Y : (f(x), g(y)) \in U \times V\} \\
&= \{(x, y) \in X \times Y : f(x) \in U,\ g(y) \in V\} \\
&= f^{-1}(U) \times g^{-1}(V).
\end{aligned}$$

Therefore,

$$\boxed{(f \times g)^{-1}(U \times V) = f^{-1}(U) \times g^{-1}(V).}$$

Step 2. Show that the preimage is open in $X \times Y$.
Since $f : X \to A$ is continuous and U is open in A, we know that

$$f^{-1}(U)$$

is open in X.

Similarly, since $g : Y \to B$ is continuous and V is open in B,

$$g^{-1}(V)$$

is open in Y.

Hence,

$$f^{-1}(U) \times g^{-1}(V)$$

is open in $X \times Y$, by the definition of the product topology.

Consequently,

$$(f \times g)^{-1}(U \times V)$$

is open in $X \times Y$.

Step 3. Conclude continuity.

We have shown that the preimage under $f \times g$ of every basic open set in $A \times B$ is open in $X \times Y$.

Since every open subset of $A \times B$ is a union of basic open sets, the preimage of every open subset of $A \times B$ is open in $X \times Y$.

Therefore,

$$\boxed{f \times g : X \times Y \to A \times B \text{ is continuous.}}$$

□